

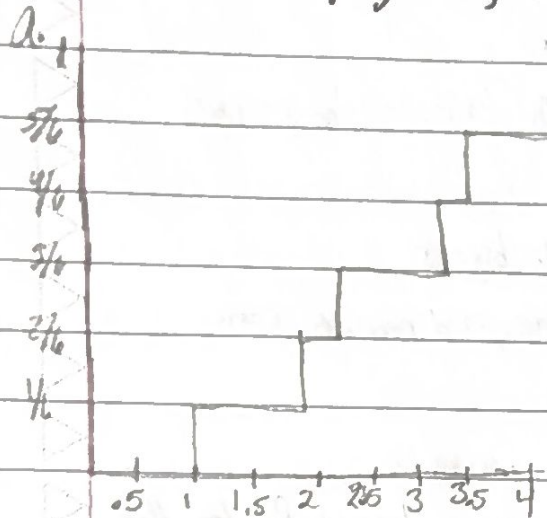
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Problems: Kernel Density Estimation

1. $[1, 1.75, 2, 3, 3.25, 4]$

$$n = 6, h = 0.2$$

$$\text{height of one box} = \frac{1}{2nh} = \frac{1}{2 \cdot 6 \cdot 0.2} = \frac{1}{2.4} \approx 0.417$$

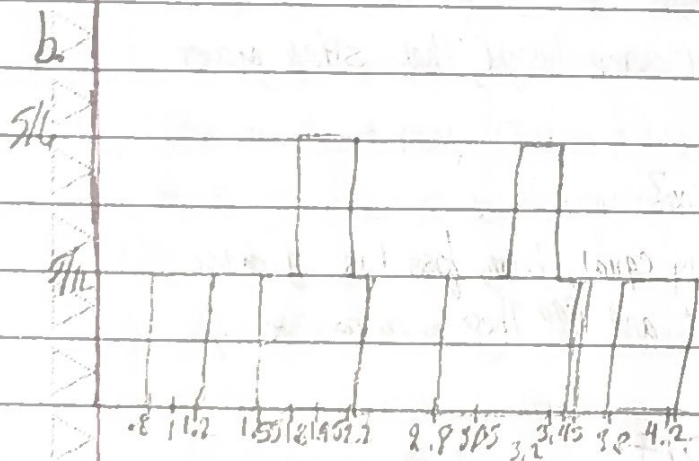


Boxes	Point	Box
	1	$[0.8, 1.2]$
	1.75	$[1.55, 1.95]$ } overlap
	2	$[1.8, 2.2]$
	3	$[2.8, 3.2]$
	3.25	$[3.05, 3.45]$ } overlap
	4	$[3.8, 4.2]$

ECDF

$$\text{mean} = (1 + 1.75 + 2 + 3 + 3.25 + 4) / 6 = 2.5$$

SD	x	$(x - 2.5)^2$	Sample SD: $\sqrt{\frac{6.125}{5}} \approx 1.107$
	1	2.25	
	1.75	0.5625	
	2	.25	
	3	.25	
	3.25	.5625	
	4	2.25	

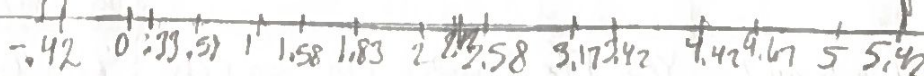


KDE, $h = 0.2$

$$h = 1.84 \cdot 1.107 \cdot 6^{1/5} \approx 1.423$$

$$\text{height} = \frac{1}{2 \cdot 6 \cdot 1.42} = 0.0587$$

Point	Box
1	$[-.42, 2.42]$
1.75	$[-.33, 3.17]$
2	$[-.58, 3.42]$
3	$[1.58, 4.42]$
3.25	$[1.83, 4.67]$
4	$[2.58, 5.42]$



KDE, $h = 1.42$

Kernel Density Estimation Problems

e. How are b and d similar or different? The bandwidth for part d is $\approx 7 \times$ larger than $h=0.2$ in part b, so the boxes overlap much more creating the much broader and smoother hill centered around 2.5. The smaller bandwidth produced 4 spiky humps because of the lack of box overlap.

2. a. $X = [-1, -0.5, 0, 0.5, 1]$, drawn from standard Normal distribution

$$f_h(0) = \frac{\# \text{ of points w/in } h \text{ of } 0}{2nh}$$

$$f_h(0) = \frac{3}{2(5)(.6)} = 0.5$$

Point

w/in .6 of 0?

-1

No

-0.5

Yes

0

Yes

.5

Yes

1

No

b. $\# [f_h(0)]$ w/ $h=0.6$ $E[f(0)] = (F(0.6) - F(-0.6)) / 2 \cdot 0.6$

using z-table for normal: $F(0.6) = 0.7257$

$$F(-0.6) = 1 - 0.7257 = 0.2743$$

$$\# [f(0)] = \frac{0.7257 - 0.2743}{1.2} \approx 0.376$$

c. True density at $x=0$ using Normal dist:

$$\text{Normal density: } f(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-x^2/2}$$

$$f(0) = \frac{1}{\sqrt{2\pi}} \approx 0.399$$

c. No, they aren't equal. Bias = $0.376 - 0.399 = -0.023$, so KDE estimates density at 0.

The expected value of $f(0)$ is the avg height of the true density from $[-.6, .6]$.

At 0, bell curve is at its peak, and averaging a peak w/ lower values produces a value $<$ peak.

$$2e. \mathbb{E}[F(0)] = \frac{(F(0.1) - F(-1))}{2 \cdot 0.1} = \frac{0.5398 - 0.4602}{0.2} \approx 0.398$$

Gap $F(0) - \mathbb{E}[F(0)] = .399 - .398 = .001$, gap nearly disappears. W/ a tiny window, density barely changes across it, so window's average is almost exactly the peak.

f. $\hat{f}_h$ is only a consistent estimator depending on how h is chosen. A fixed h is not consistent bc w/o changing window size, bias can't be removed. When h shrinks as n grows, # of observations in each window still grows so variance also goes to 0.

g. The main issue is the bias variance tradeoff. With small h , bias is low but variance is high bc few points in each window = noisy estimate. A large h lowers variance but increases bias bc many points in a window gives a smooth estimate that flattens peaks and flattens tails.

3a. $X = [5, 7, 7, 8, 10]$, $h = 1.5$

x	Distances to $[5, 7, 7, 8, 10]$	Count < 1.5	$f(x) \frac{\# < 1.5}{2nh} = \frac{\# < 1.5}{2(5)(1.5)}$
6	1, 1, 1, 2, 4	3	$\frac{3}{15} = 0.2$
6.5	1.5, .5, .5, 1.5, 2.5	2	$\frac{2}{15} = 0.133$
7	2, 0, 0, 1, 3	3	$\frac{3}{15} = 0.2$
8	3, 1, 1, 0, 2	3	$\frac{3}{15} = 0.2$
8.5	3.5, 1.5, 1.5, .5, 1.5	1	$\frac{1}{15} = 0.067$

3b Gaussian kernel: $f(x) = \frac{1}{nh} \sum \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}(\frac{x_i - x}{h})^2}$

$\frac{1}{5 \cdot 1.5 \cdot 2.07} \approx 0.0532$

$f(x) = 0.0532 \times \text{sum of points}$

$x = 6$

x_i	Distance	$e^{-\frac{(\text{dist})^2}{2}}$
5	1	.801
7	1	.801
7	1	.801
8	2	.135
10	4	.002

$x = 6.5$

x_i	Distance	$e^{-\frac{(\text{dist})^2}{2}}$
5	1.5	.882
7	1.5	.882
7	1.5	.882
8	1.5	.325
10	3.5	.002

Sum $2.842 \cdot 0.0532 = 0.151$

Sum $2.913 \cdot 0.0532 = 0.169$

$x = 7$

x_i	Distance	$e^{-\frac{(\text{dist})^2}{2}}$
5	2	.135
7	0	1
7	0	1
8	1	.801
10	3	.011

$x = 8$

x_i	Distance	$e^{-\frac{(\text{dist})^2}{2}}$
5	3	.011
7	1	.801
7	1	.801
8	0	1
10	2	.135

Sum $2.947 \cdot 0.0532 = 0.178$

Sum $3.024 \cdot 0.0532 = 0.167$

$x = 8.5$

x_i	Distance	$e^{-\frac{(\text{dist})^2}{2}}$
5	3.5	.002
7	1.5	.325
7	1.5	.325
8	.5	.882
10	1.5	.325

Sum $1.859 \cdot 0.0532 = 0.151$

3c. x	Uniform	Gaussian
6	.200	.151
6.5	.185	.169
7	.200	.178
8	.200	.167
8.5	.067	.151

Biggest jump is from 8 to 8.5. Uniform KDE drops from .2 to .067 while Gaussian rises from .167 to .151.

The uniform kernel gives each point a weight of $1/(2h)$ within h of x and 0 otherwise. Moving from 8 to 8.5 puts both observations of 7 at 1.5 from x , so they drop out and the count goes from 3 to 1. Gaussian Kernel's weight declines smoothly w/ distance, so those two points have their weights dropped instead of being removed entirely. No obs. switches abruptly on or off w/ Gaussian so the KDE changes continuously.

1. The uniform kernel is simpler and easier to interpret (estimate at x is fraction of obs w/in h of x $\div$ by $2h$) and also easier to compute by hand. However, it produces blocky discontinuous estimates.

The Gaussian kernel gives a smooth estimate by downweighting distant observations gradually.

Downside is all observations influence estimate everywhere, so it may put density where the variable can't go.